Problem Definition

Companies involved in logistics and also delivery drivers experience delivery delays, damage/road accidents and an increase operational costs due to road conditions.

Existing navigation systems installed in today's world primarily consider distance and estimated travel time without sufficiently considering multiple factors that can affect the safety of a route.

Factors including traffic volume, recent road accidents, severe weather, restricted roads, potholes on roads and other road changing conditions. This results to delivery drivers being guided onto routes that are short in distance but jammed with traffic, routes that are also exposed to extreme weather conditions or restricted roads causing longer delivery times, increased fuel consumption and unreliable deliveries.

This challenge is important for AI solutions for industries because it identifies an operational issue in the logistics industry for which current technology falls short.

Addressing this issue is beneficial to the community and to the municipality. Efficient logistics vehicle routing reduces community congestion and road safety risk for municipal infrastructure. Emergency response times will be faster and vehicle idle times that cause harmful emissions will be reduced. Finally, there will be improved safety and environmental quality for citizens.